

LH-DETR: A Lightweight Hybrid Architecture for End-to-End Object Detection in UAV Images

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Problem and Goal

UAV object detection must handle small targets, dense scenes, cluttered backgrounds, and onboard compute limits. Existing DETR-style detectors remove anchors and NMS, but still depend on expensive high-resolution features and often miss detail-rich small objects.

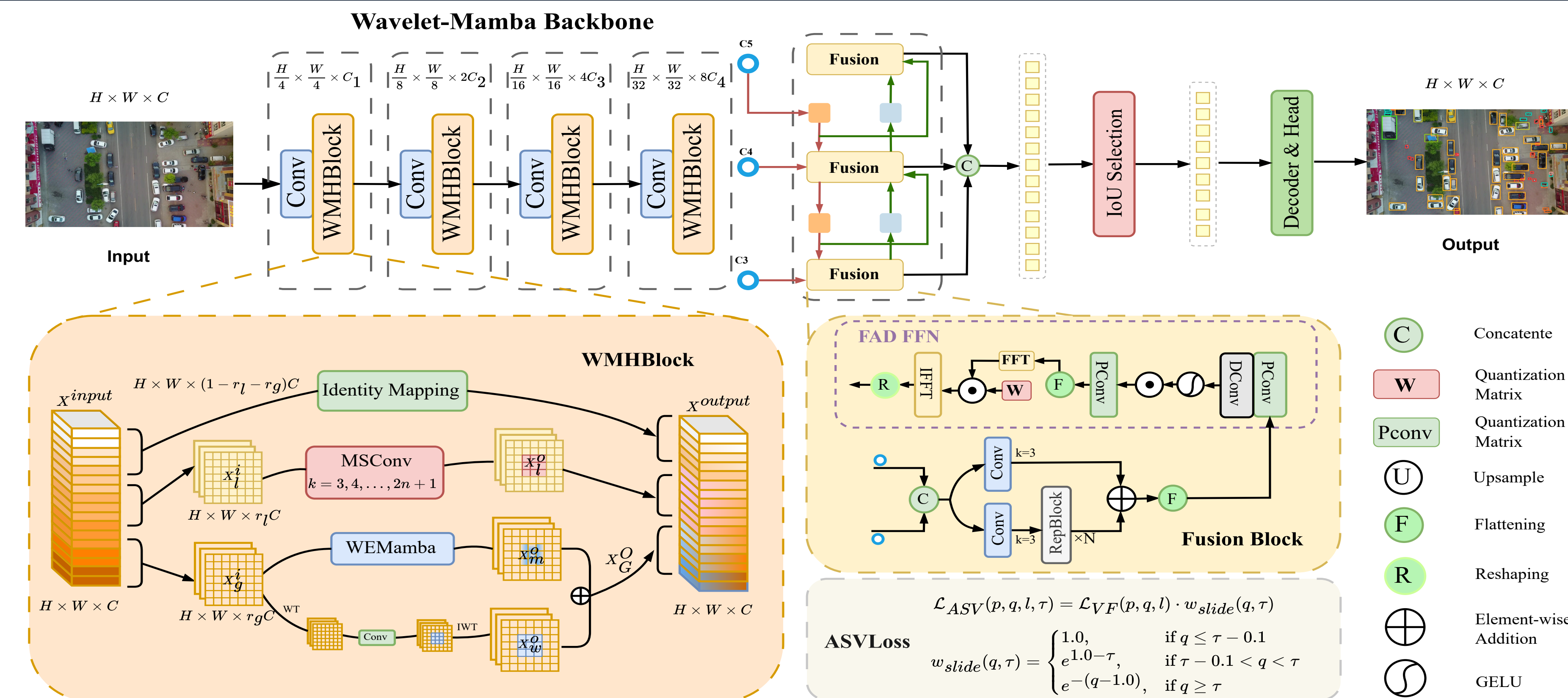
Goal: build an end-to-end detector that preserves DETR's clean prediction pipeline while improving the accuracy-efficiency trade-off for aerial images.

Core Contributions

- **Wavelet-Mamba Hybrid Block (WMHB):** splits channels into global, local, and identity paths; combines wavelet decomposition with Mamba state-space modeling for long-range context and multi-scale detail.
- **Frequency-Aware Dynamic FFN (FAD-FFN):** applies local 8×8 RFFT filtering to amplify high-frequency edges and textures that matter for small targets, while reducing standard FFN parameters from $8C^2$ to $6C^2$.
- **AutoSliding Varifocal Loss (ASVLoss):** uses an adaptive IoU reference point $\tau = \max(\text{auto_iou}, 0.2)$ to shift training focus from medium-quality predictions to high-quality predictions.

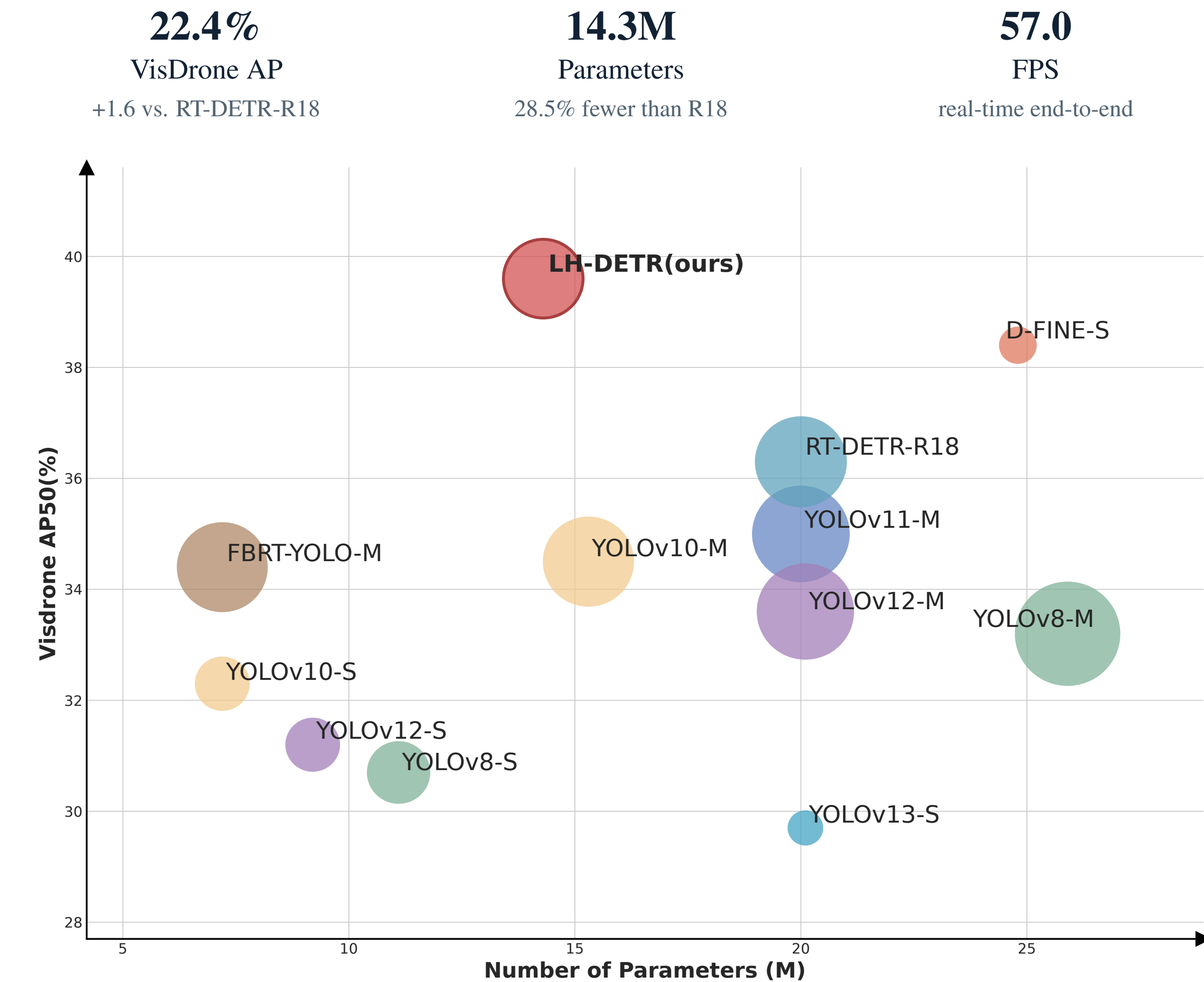
Module relationship: WMHB improves representation efficiency, FAD-FFN strengthens detail sensitivity, and ASVLoss improves ranking quality during training. The modules address complementary bottlenecks rather than repeating the same operation.

Architecture Overview



Reading guide. LH-DETR keeps the end-to-end RT-DETR detection pipeline, but replaces costly feature processing with a lighter hybrid representation. WMHB routes channels into global, local, and identity paths; FAD-FFN injects frequency-aware enhancement into the encoder; ASVLoss guides optimization toward progressively higher-quality predictions. The intended effect is a detector that is easier to deploy on UAV perception systems: global context is retained for crowded scenes, high-frequency details are preserved for small targets, and the final head remains NMS-free.

Headline Results



Method Snapshot

WMHB channel routing

$$\mathbf{X} = [\mathbf{X}_{global}, \mathbf{X}_{local}, \mathbf{X}_{identity}]$$

Global channels use DWT + Mamba + IDWT; local channels use multi-kernel depthwise convolution; identity channels preserve information flow. Outputs are concatenated and projected with a residual connection.

ASVLoss weighting

$$w(q, \tau) = \begin{cases} 1.0, & q \leq \tau - 0.1 \\ e^{1-\tau}, & \tau - 0.1 < q < \tau \\ e^{-(q-1)}, & q \geq \tau \end{cases}$$

What Changes in the Detector

1. Global path

Wavelet low-frequency components are processed by Mamba selective scanning, providing linear-complexity long-range dependency modeling.

2. Local path

Multi-scale depthwise convolutions with 3×3 , 5×5 , and 7×7 kernels capture small-object boundaries and textures.

3. Frequency FFN

Local FFT filtering selectively enhances frequency components and fuses them back with lightweight spatial processing.

Design principle: allocate computation only where aerial detection needs it most: long-range context, multi-scale local detail, and frequency cues for small dense objects.

VisDrone Comparison

Model	Params	FLOPs	FPS	AP	AP ₅₀
RT-DETR-R18	20.0M	60.0G	55.6	20.8	36.3
RT-DETR-R50	42.0M	136G	53.5	22.2	39.1
D-FINE-M	56.3M	19.1G	-	24.2	41.7
LH-DETR	14.3M	44.9G	57.0	22.4	39.6

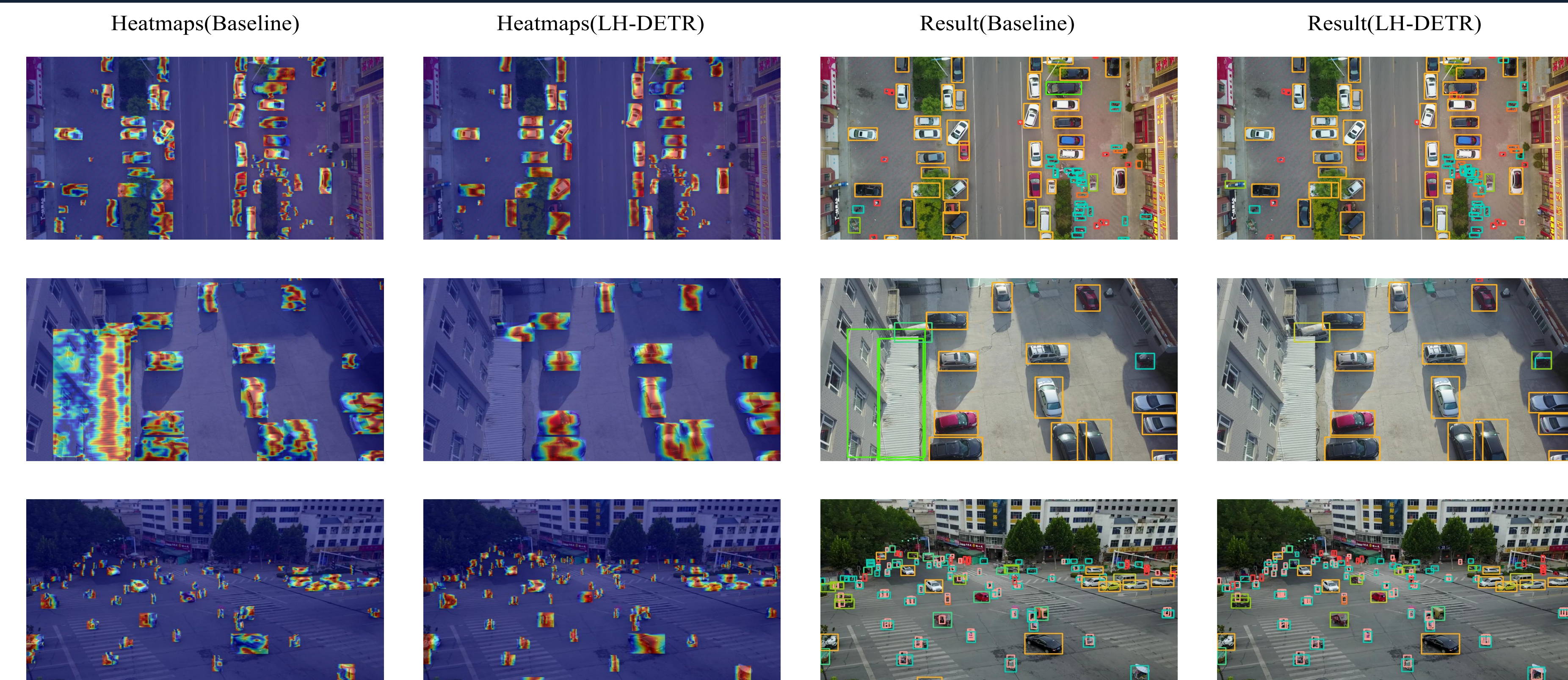
LH-DETR improves AP by 1.6 points over RT-DETR-R18 while reducing parameters by 28.5% and FLOPs by 25.2%.

Training Setup

Models are trained for 300 epochs with SGD on an NVIDIA RTX 4090 and evaluated for inference speed on an RTX 3090. Input resolution is 640×640 unless otherwise specified.

This controlled setup keeps the comparison focused on architectural changes rather than differences in image scale or optimization schedule.

Qualitative Attention



Conclusion and Deployment

LH-DETR advances real-time DETR-style aerial detection by reducing redundant computation while preserving global context and fine-grained high-frequency cues.

- **Practical value:** stronger AP/FPS/parameter balance for UAV imagery, with a lighter profile than RT-DETR-R18.
- **Scene robustness:** better localization in dense traffic, parking, and pedestrian scenes where small objects occupy only a few pixels.
- **Deployment path:** optimize DWT and related non-standard operators for TensorRT, then validate latency, memory use, and stability on embedded UAV hardware.

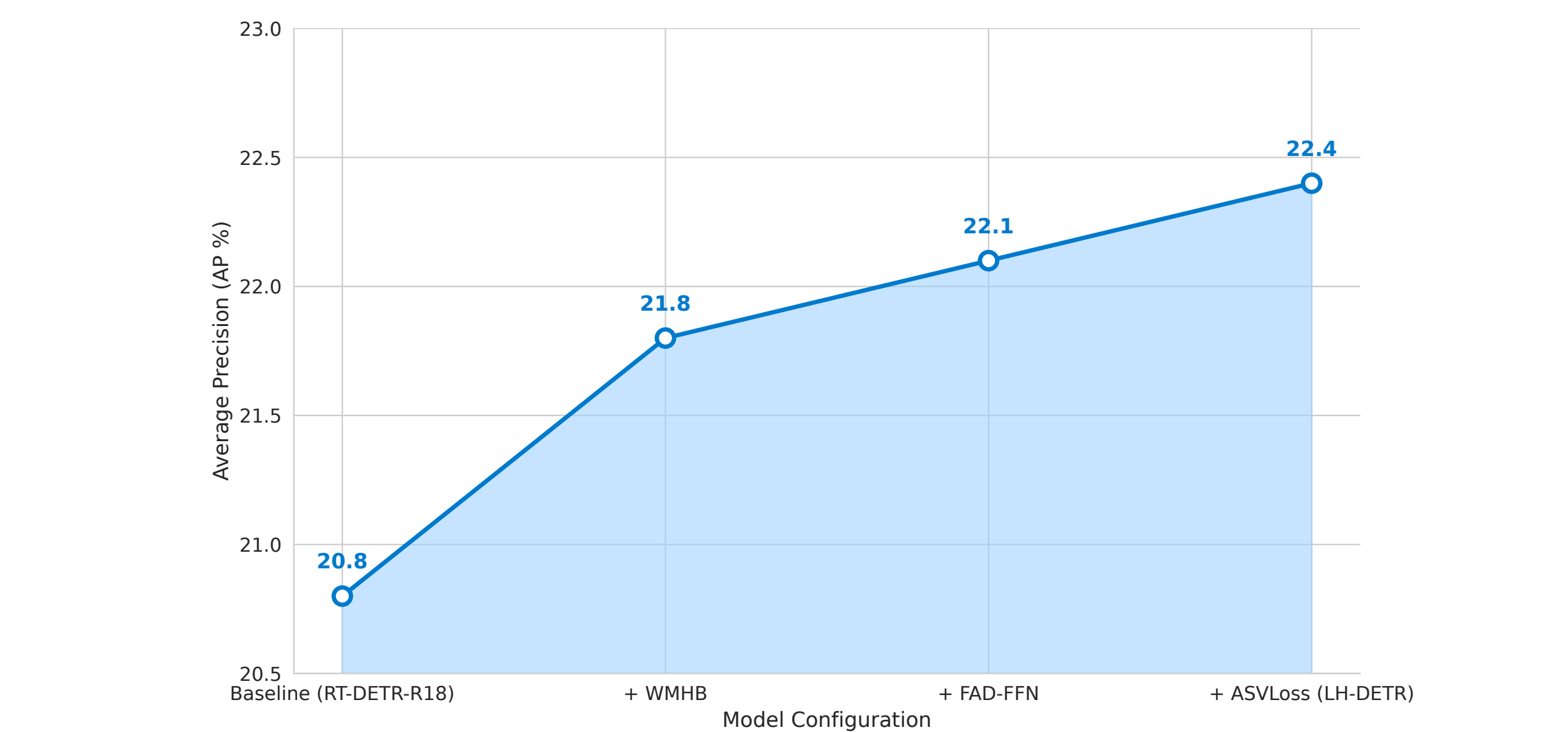
The method keeps the end-to-end detector interface.

Generalization Across UAV Datasets

Dataset / Model	Params	FPS	AP	AP ₅₀
UAVVaste RT-DETR-R18	20.0M	55.6	47.7	78.6
UAVVaste LH-DETR	14.3M	57.0	49.0	79.7
UAV-PDD RT-DETR-R18	20.0M	55.6	31.8	63.4
UAV-PDD LH-DETR	14.3M	57.0	33.0	64.3

Consistent AP gains on UAVVaste and UAV-PDD indicate that the improvements are not limited to VisDrone traffic scenes.

Ablation



The full model reaches 22.4 AP and 39.6 AP₅₀. WMHB is the main efficiency driver; FAD-FFN and ASVLoss provide cumulative accuracy gains without changing the deployment profile of the final lightweight detector.